

Topology-Risk Active Inspection for Wire Harnesses: Public Cable Defects, Paired Budget Tests, and Executable Robot Scans

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Abstract—Robot inspection of branched wire harnesses couples anomaly detection with the choice of where to spend a limited scan budget. We present InspectBot, a hybrid benchmark grounded in two public real-image datasets and executable 3-D robot scans. A spatial Gaussian model trained only on normal MVTec cable images reaches 92.2% image and 95.7% pixel AUROC. On independent stripped-wire data, a spatial nearest-neighbor reference reaches 81.5% AUROC but only 41.3% recall at a normal-only threshold. Occlusion plus darkening reduces anomaly recall to 1.1%, exposing a perception floor that no scan policy repairs. Under a disjoint 2,000-scenario protocol, fixed topology-risk ordering improves criticality-weighted recall over geometry coverage by 9.4, 7.8, 7.7, and 4.7 percentage points at budgets 4, 6, 8, and 10; all remain significant after a 30-test Holm correction. The advantage disappears at full coverage and reverses under naive revisits. Direct MuJoCo execution completes all 320 runs with sub-0.97-mm field-of-view and sub-1.50-mm visual-IK error. The evidence supports risk-aware budget allocation, not learned topology, hardware safety, or factory performance.

Note to Practitioners—Wire-harness inspection stations often have less camera time than is needed to view every terminal, branch, label, and clip at equal quality. This paper tests a simple operational principle: spend the limited scan budget first at sites whose process role makes a missed defect costly, but stop trusting the planner when image quality is too poor. On public cable and stripped-wire images, the experiments show that adverse views can impose a perception floor that no view ordering policy repairs. Under cleaner views, topology-risk ordering improves criticality-weighted recall at partial coverage, but the advantage disappears at full coverage and reverses under poorly designed revisits. The current robot is a digital twin, so the reported rates are not factory escape rates. Before deployment, practitioners must calibrate the actual cameras and fixtures, measure cycle time, and validate the policy across suppliers, lighting and real automotive harness defects.

Index Terms—Active perception, anomaly detection, industrial inspection, robot vision, wire-harness assembly.

I. Introduction

Wire-harness inspection is not uniform search. Damage near a branch junction, high-current path, or safety-critical connector can carry greater engineering risk than an equally visible defect on a redundant segment. At the same time, the camera and anomaly detector may

be least reliable at occluded or reflective locations. An inspection robot must therefore decide not merely whether an image is anomalous, but which site should be observed next and whether revisiting it is preferable to covering an unobserved site.

Industrial anomaly detection has strong controlled benchmarks and capable one-class methods [3], [4], [7]. CableInspect-AD shows that expert cable anomalies remain difficult outside controlled factories [1]. Wire-harness reviews similarly identify robustness, practical validation, and exploitation of intrinsic harness structure as open problems [10]. Existing evidence, however, rarely couples a real-image detector to paired resource-allocation tests and executable scan motions. Merely rendering a robot beside a cable does not close that gap.

InspectBot narrows the claim to a falsifiable question: given frozen detector scores and a fixed engineering-risk map, does topology-aware allocation recover more risk-weighted anomaly mass than non-topological policies at the same number of observations? Its contributions are:

- 1) two integrity-audited public-data experiments with normal-only fitting and threshold calibration, including an independent stripped-wire refit;
- 2) a 600-row real-image score cache under four deterministic appearances, exposing both a useful clean-view regime and a severe adverse-view failure;
- 3) paired stress and deployment protocols covering 126,000 active-policy rows, exact frozen seeds, bootstrap intervals, multiplicity-corrected tests, an explicit shuffled-risk control, and a corrected offline oracle;
- 4) a direct MuJoCo execution audit that preserves all allocation outcomes while testing bounded 3-D motion to the 12 inspection sites; and
- 5) a machine-audited claim boundary separating real images, synthetic appearance transforms, simulated robot motion, and absent hardware evidence.

II. Related Work

a) Industrial anomaly detection.: MVTec AD established a controlled multi-category one-class benchmark [3]. PaDiM fits a Gaussian distribution at each pretrained feature location [4]; PatchCore instead retains a coreset of nominal patch features [7]. We use these as transparent method families, not as a novelty claim. Our compact

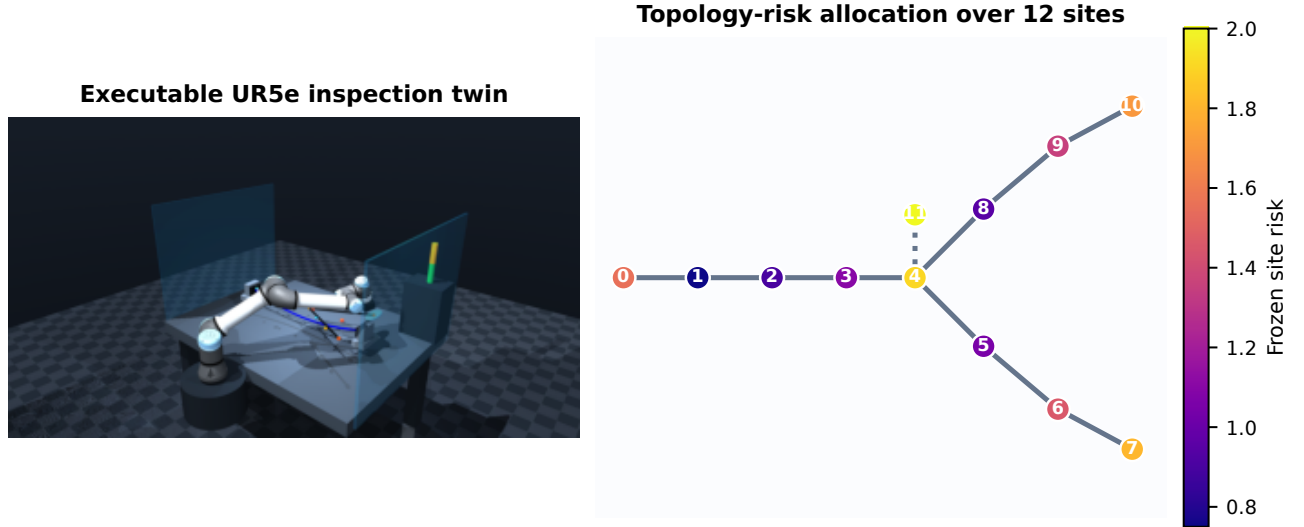


Fig. 1. InspectBot separates allocation evidence from execution evidence. Left: the UR5e visual twin executes bounded scanner motions in MuJoCo. Right: 12 frozen inspection sites and their engineering-risk weights; the displaced dotted node is a process checkpoint physically colocated with the junction. Detector scores come from held-out real images, not from the rendered camera.

spatial Gaussian keeps 64 deterministic channels from a frozen ResNet-18, while the independent validation also includes a position-wise nearest-neighbor reference.

CableInspect-AD contains expert-annotated high-resolution power-line cable images with multiple real-world anomaly types [1]. We audited its official code but could not obtain the dataset from its official endpoints without an access request; consequently no CableInspect number appears in our tables. The FAU/FAPS Stripped Wire Dataset contains good, cut-strand, and pulled-strand images [8]. Its associated PB-IAD study uses semantic foundation-model prompts and a PatchCore comparison [5]. That comparison selects a threshold using labeled test data, whereas every InspectBot threshold is selected from normal calibration images only; thresholded F1 values are therefore not directly comparable.

b) Wire-harness perception and active inspection.: Multi-branch 3-D vision has been validated in a real dual-robot assembly setting [6], which is stronger hardware evidence than ours. We address a different decision: distributing observations after the sites and a risk map are known. This follows the active-perception view that sensing actions should be selected jointly with task goals [2]. The risk map here is engineering input, not learned from labels. A shuffled-risk policy is therefore essential: it tests whether improvements require the correct site-risk binding rather than merely a nonuniform scan order.

III. Public-Data Perception

A. Data integrity and splits

The MVTec cable audit fixes the transport mirror at commit c75b39616f84 and the extracted tree at SHA-256 prefix 5ecf3b6b9fee. It contains 224 normal training images, 58 normal test images, 92 anomalous test images across eight defect types, and 92 masks. A seed-20260827

permutation assigns 180 normals to fitting and 44 to threshold calibration. The public test set is untouched until evaluation.

The Stripped Wire archive is fixed to DOI 10.5281/zenodo.16686806, version v2, MD5 9e1944f1b7c9bc9e7db433f7bdcf2694. Its 200 released PatchCore normals are split 160/40 for fitting/calibration; all 133 good, 68 cut-strand, and 99 pulled-strand test images remain held out. Exact-file hashing finds no train-test duplicate. Three optional VLM references are excluded. Aspect ratio is preserved and padded to 224×224 , avoiding a test-label-dependent crop.

B. Detector and baselines

Frozen ImageNet ResNet-18 features from layers 1–3 are aligned to a 28×28 grid. A seed-fixed subset of 64 channels forms feature x_{ijp} for normal image j at spatial position p . The spatial Gaussian score is

$$a_p(x) = \sqrt{(x_p - \mu_p)^\top (\Sigma_p + 10^{-2}I)^{-1} (x_p - \mu_p)}, \quad (1)$$

and the image score averages the top 1% of spatial values. The strong spatial nearest-neighbor reference replaces the Gaussian distance with the closest normal feature at the same position. RGB statistics and globally pooled ResNet features, each scored with Ledoit–Wolf Mahalanobis distance, are weaker controls. Each method’s threshold is its own 99th percentile on normal calibration scores. No anomalous image selects features, preprocessing, thresholds, or models.

C. Perception results

On MVTec cable, the spatial Gaussian reaches 92.2% image AUROC (stratified image-bootstrap 95% CI 87.3–96.1) and 94.7% AUPR. Its normal-calibrated operating

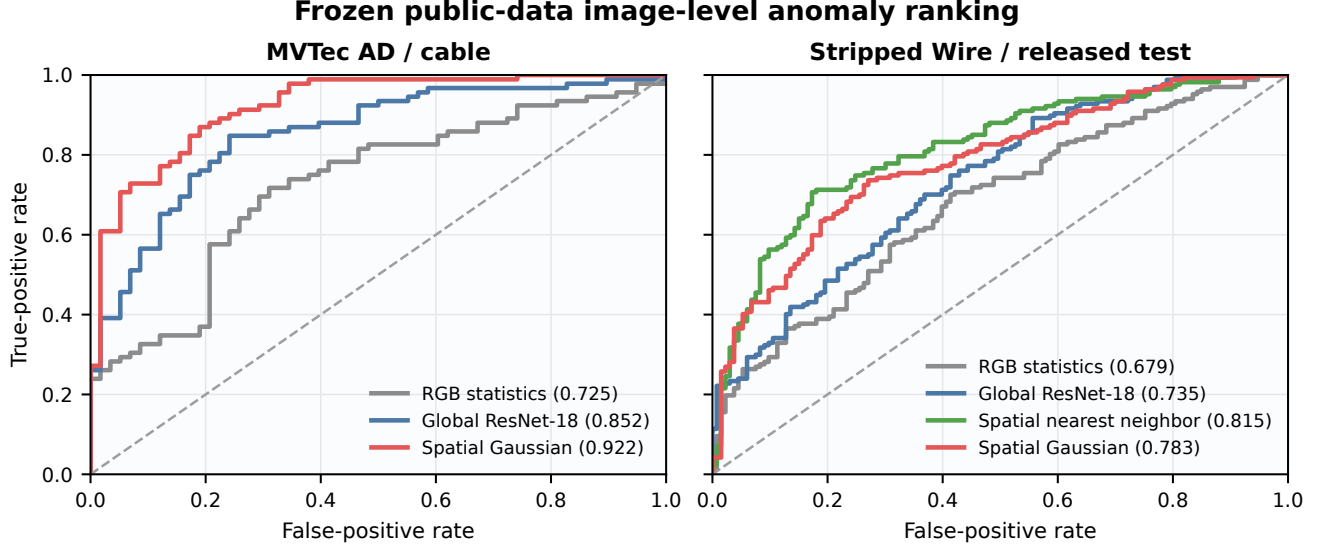


Fig. 2. Image-level ROC curves recomputed from frozen per-image CSV files. MVTec favors the spatial Gaussian model; the independent wire-end task favors the spatial nearest-neighbor reference.

TABLE I

MVTec cable test results (58 normal/92 anomaly). Values are percent; the decision threshold uses 44 normal calibration images only.

Method	AUROC	AUPR	Recall	F1
RGB statistics	72.5	82.0	56.5	66.7
Global ResNet-18	85.2	90.4	59.8	71.4
Spatial Gaussian	92.2	94.7	65.2	77.4

TABLE II

Independent Stripped Wire refit (133 normal/167 anomaly). Spatial NN is an exploratory strong reference, not the prespecified candidate.

Method	AUROC	AUPR	Recall	F1
RGB statistics	67.9	73.3	20.4	33.0
Global ResNet-18	73.5	78.2	34.1	47.9
Spatial nearest neighbor	81.5	83.9	41.3	56.3
Spatial Gaussian	78.3	81.8	36.5	52.4

point gives 65.2% recall, 95.2% precision, and 5.2% false-positive rate. Pixel AUROC is 95.7%, but pixel AUPR is 48.2% and F1 is 52.5%; localization is useful but far from solved.

On Stripped Wire, spatial nearest neighbor is best at 81.5% AUROC (76.5–86.1) and 83.9% AUPR. Spatial Gaussian reaches 78.3% AUROC, exceeding RGB statistics by 10.3 points (paired bootstrap CI 4.7–16.0) and global features by 4.8 points (0.2–9.4), but it is 3.2 points below spatial NN (–5.4 to –1.1). The strict spatial-NN threshold yields only 41.3% recall. Its per-defect AUROC is 71.1% on cut strands and 88.6% on pulled strands, exposing the harder failure mode. We therefore do not claim method superiority or production readiness.

TABLE III

Frozen detector robustness by appearance (150 held-out images). Each threshold is calibrated on transformed normal training images.

View	Transform	AUROC	Recall	FPR
0	clean center crop	92.2	65.2	5.2
1	darkened and blurred	92.6	73.9	1.7
2	rotated and contrast shifted	62.9	10.9	1.7
3	occluded and darkened	55.2	1.1	0.0

IV. Hybrid Active Inspection

A. Real-image score bridge

Every held-out MVTec image is scored under four deterministic appearances: clean; dark plus blur; rotation plus contrast; and deterministic occlusion plus darkening. Each appearance v is independently calibrated on the same 44 normal images. We store

$$q_{iv} = \frac{s_{iv} - \tau_v}{1.4826 \text{ MAD}_v}, \quad (2)$$

where τ_v is the 99th-percentile normal threshold. A site's prediction is the sign of the mean of all observed q_{iv} . These transforms are controlled stress tests, not physical views or camera trajectories.

The clean and dark-blur appearances retain 92.2% and 92.6% AUROC. Rotation plus contrast falls to 62.9%; occlusion plus darkening falls to 55.2% AUROC and 1.1% recall. This observation motivates two frozen protocols rather than deletion of the failed setting.

B. Sites, outcome, and policies

Each paired scenario assigns a held-out real image to each of 12 sites on a branched topology, with anomaly probability 0.30 conditioned on at least one normal and

Deterministic class-level qualitative audit (no best-case selection)

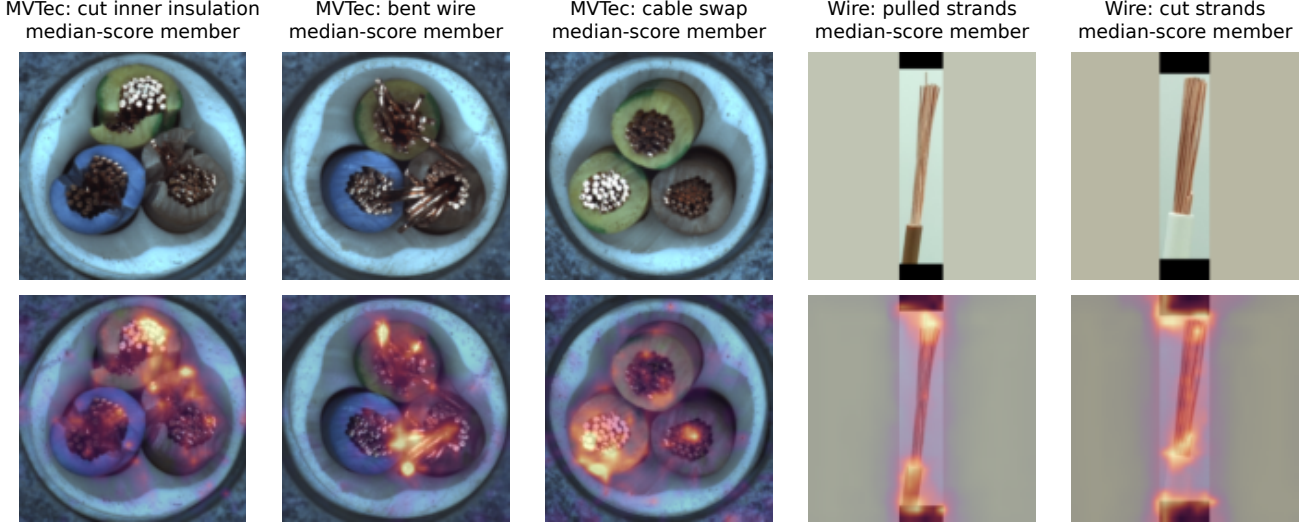


Fig. 3. Deterministic qualitative audit. The three largest MVtec defect classes by test count and both Stripped Wire defect classes are shown; within each class the image nearest its median anomaly score is selected. Per-image robust color scaling aids localization display but is not used for decisions. Boundary responses on wire ends reveal a concrete localization weakness.

one anomalous site. Defect type is sampled uniformly before an image is selected within type. Site risk r_i and defect-severity coefficient d_i are fixed before the run. For true anomaly indicator y_i and final prediction $\hat{y}_i$, the primary metric is

$$\text{CWR} = \frac{\sum_i r_i d_i y_i \hat{y}_i}{\sum_i r_i d_i y_i}. \quad (3)$$

This metric intentionally values critical sites. Ordinary recall, precision, F1, false-positive rate, coverage, revisits, travel, and complete critical success are secondary outcomes.

The seven policies are raster order, random scanning, nearest-unvisited geometry coverage, coverage-first uncertainty revisit, topology risk plus uncertainty, a shuffled-risk negative control, and an offline label/score-aware oracle. The topology policy selects unvisited site i approximately by $r_i/(1 + \text{travel}_i)$ and, after coverage, combines risk with score uncertainty. It is a deterministic engineering heuristic, not a learned planner. The oracle solves an exact multiple-choice knapsack over per-site view prefixes and is descriptive only.

C. Frozen paired protocols and statistics

Stress-v1 uses 1,000 scenarios (seeds 77,000,000–77,000,999), budgets $\{4, 6, 8, 10, 12, 16\}$, and the appearance order adverse-dark, dark-blur, rotate-contrast, clean. Deployment-v2 was written after retaining that result; it changes only to clean-first order and uses 2,000 disjoint scenarios (seeds 78,000,000–78,001,999). Detector scores, topology, risks, severities, policy equations, thresholds, budgets, and tests remain fixed.

At each budget we compare topology risk with five non-oracle baselines using paired, one-sided Wilcoxon

signed-rank tests on CWR. All 30 p -values are Holm adjusted together. Scenario-paired bootstrap intervals use 10,000 resamples. Budgets 4–10 are confirmatory under-coverage regimes; budget 12 is a prespecified coverage equality check and budget 16 is exploratory. An initial greedy implementation mislabeled as “oracle” was not a true upper bound. We retain that run, correct only the descriptive oracle, and audit that all 72,000 non-oracle rows and the complete paired-comparison CSV remain unchanged.

V. Active-Inspection Results

a) The adverse-first protocol fails informatively.: Topology risk reaches only 0.5% CWR at budget 4, and no non-oracle policy shows a stable advantage through budget 12: the first site observation has only 1.1% anomaly recall. Random revisits occasionally encounter a less damaging transform earlier. Allocation cannot compensate for an arbitrarily poor sensing regime.

b) Topology matters under clean partial coverage.: In deployment-v2, topology risk reaches 26.8% CWR at budget 4, versus 17.4% for geometry, a 9.4-point paired gain (95% CI 8.0–10.9, Holm-adjusted p value 8.08×10^{-38}). The gains at budgets 6, 8, and 10 are 7.8, 7.7, and 4.7 points. At all four confirmatory budgets topology also exceeds raster, random, uncertainty, and shuffled-risk after the same 30-test correction. The shuffled control supports the interpretation that correct site-risk binding, rather than an arbitrary nonuniform order, drives the primary metric.

c) Coverage and revisits bound the claim.: At budget 12, all coverage-first policies inspect every site once and tie at 64.8% CWR. At budget 16, topology risk falls to 61.2%, below geometry’s 63.6% and uncertainty’s 62.6%.

Perception quality bounds the value of active view planning

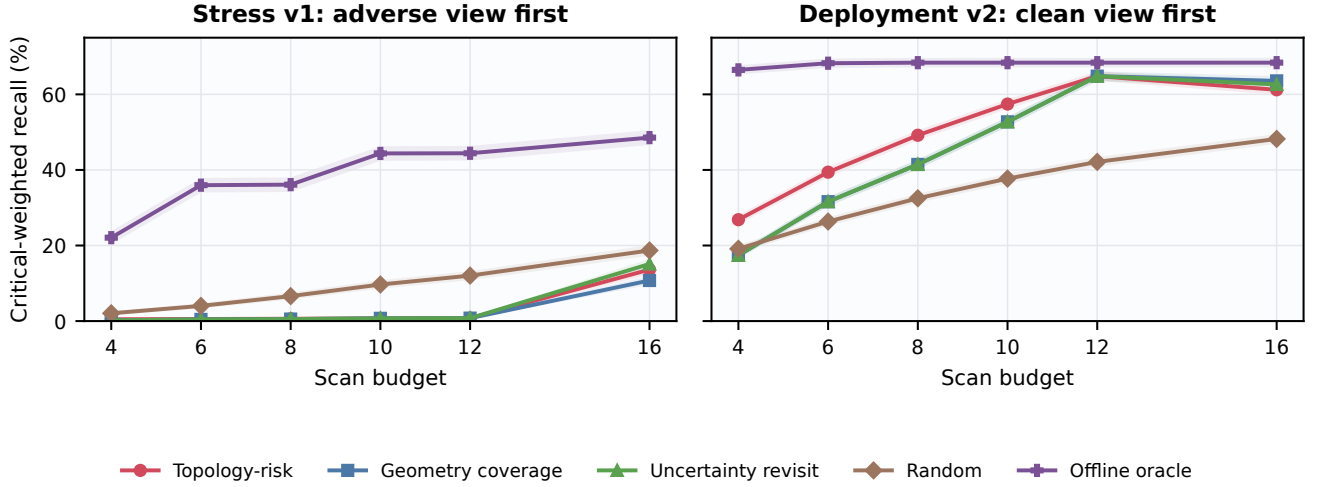


Fig. 4. Criticality-weighted recall versus scan budget on shared 0–75% axes. Stress-v1 is a retained negative result: adverse-first observations impose a perception floor. Clean-first deployment-v2 reveals a topology-risk allocation advantage under partial coverage, followed by equality and a late-revisit reversal. Bands are scenario-bootstrap 95% intervals.

TABLE IV

Deployment-v2 criticality-weighted recall in percent. Differences are paired topology minus geometry. Holm p values belong to the conservative family of all 30 topology-versus-baseline comparisons; budget 16 used a one-sided superiority alternative and is exploratory.

Budget	Topology	Geometry	Δ [95% CI]	Holm p
4	26.8	17.4	+9.4 [+8.0,+10.9]	8.08×10^{-38}
6	39.4	31.6	+7.8 [+6.5,+9.2]	4.29×10^{-30}
8	49.2	41.5	+7.7 [+6.2,+9.3]	6.33×10^{-28}
10	57.4	52.7	+4.7 [+3.6,+5.8]	2.08×10^{-17}
12	64.8	64.8	+0.0 [+0.0,+0.0]	1.00×10^0
16	61.2	63.6	-2.4 [-3.1,-1.7]	1.00×10^0

Averaging later degraded-view scores can turn a previously detected anomaly negative; risk-weighted revisiting then harms the same high-value sites it tries to protect. The corrected offline oracle uses an average of 2.64 scans and reaches 66.5% CWR at budget 4, 68.2% at 6, and 68.4% from budget 8 onward. The gap suggests a future learned stopping rule, but oracle knowledge cannot be deployed.

VI. Direct MuJoCo Audit

The direct audit uses seeds 78,000,100–78,000,119, four executable policies, and budgets 4, 8, 12, and 16. At each decision, the current 3-D inspection-site pose is read from MuJoCo [9]. The scanner moves in increments no larger than 25 mm, with four physics steps per increment, and a scan is accepted only within 3 mm of the site. A detailed UR5e plus dual-lens head follows the task-space scanner through inverse kinematics. The arm mesh is collision-disabled so it cannot silently perturb the cable benchmark.

All 320 runs reach every prescribed site. Maximum accepted field-of-view error is 0.97 mm and maximum

TABLE V

Direct execution diagnostics pooled over four budgets per policy. Travel is world-space scanner-path length.

Policy	Runs	Reach	Max IK [mm]	Travel [m]
Topology-risk	80	100%	1.49	2.10
Geometry coverage	80	100%	1.49	1.07
Uncertainty revisit	80	100%	1.49	1.22
Random	80	100%	1.50	3.29

visual IK error is 1.50 mm; every simulation state remains finite. For identical seed, policy, and budget, 14 shared perception/allocation fields exactly match deployment-v2. This is deliberate: the new evidence is executable 3-D motion, not a second performance sample. The detector scores are replayed from public images rather than computed from MuJoCo renders, and neither collision forces nor real camera calibration is validated.

VII. Limitations

The primary metric multiplies an engineered risk map by defect severity, and the topology policy receives that same risk map. Paired geometry, uncertainty, random, raster, and shuffled-risk controls establish resource-allocation value, but they do not show that risk was learned or optimally chosen. An actual factory must derive weights through traceable hazard analysis and test their sensitivity.

The active benchmark repeatedly samples a finite 150-image MVTec test pool. Intervals quantify scenario resampling, not unseen cables, suppliers, cameras, or factories. Deterministic transforms are appearance perturbations rather than robot viewpoints; site localization, autofocus, exposure control, depth, and temporal registration are absent. Stripped Wire is an independent refit, not zero-shot domain transfer, and both public datasets show controlled cable cross-sections or single wire ends rather than a complete automotive harness.

Only one frozen backbone/channel seed is evaluated. The study does not claim a state-of-the-art anomaly detector, an optimal active policy, or superiority over CableInspect-AD and PB-IAD. Deployment-v2 was designed after observing the stress-v1 perception floor, although its new protocol and disjoint seeds were frozen before execution. Reporting both versions is mandatory.

Finally, MuJoCo validates bounded task-space reachability with a visual IK twin. It does not validate joint control, collision avoidance, cable-robot contact, force limits, camera measurements, timing, safety interlocks, maintenance, or production economics. Hardware experiments and a prospective cross-factory protocol are required before any deployment claim.

VIII. Ethics and Safety

The used datasets contain industrial objects and no personal data. MVTec cable is CC BY-NC-SA 4.0; Stripped Wire v2 is CC BY 4.0. A release should distribute download instructions and hashes rather than silently relicense source images. Automated inspection may reduce repetitive visual work and improve traceability, but it may also displace tasks or create automation bias. A missed electrical defect can be safety-critical. The present detector and simulator must not be used to waive human inspection or safety validation.

IX. Reproducibility

The research package includes source URLs, transport revision, archive and tree hashes, fit/calibration manifests, checkpoints, all per-image predictions and maps, the 600-row view cache, per-scenario active CSVs, the retained oracle bug, frozen protocols, MJCF, per-run motion traces, plot code, and a one-command paper build. An independent audit recomputes public counts, file hashes, AUROC/AUPR, threshold outcomes, every active aggregate, oracle upper-bound behavior, all non-oracle correction rows, MuJoCo tolerances, and cross-system outcome identity. All 787 checks must pass before the release PDF is copied.

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